Weather, Not Climate: Emotional Context Dynamics in Transformer KV-Cache Geometry

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Abstract

Single-turn KV-cache keys carry the emotion of the text being processed: prior work, corrected in October 2026, decodes from keys the emotion a generator was asked to write a story in, at 12.3× chance. That decode is not significantly above a text (TF-IDF) baseline on the same stories (0.410 against 0.370; Holm-corrected $p = 0.071$), and its margin depends on a residualization fitted on each test fold (0.348, exploratory, when fitted on the training fold). The method that work describes is not above text (0.358 against 0.382, at 10 folds). We ask whether an emotion signal in the keys *accumulates* across conversation turns—whether the model builds up a reading of the conversation's emotional tone that grows with interaction (the tone of scripted messages, not a user's state). Four experiments reveal a nuanced answer: emotion does not accumulate cumulatively, but it is not stateless either.

A dose-response experiment (1–8 emotional turns) shows that cheerful/hostile discrimination is *maintained* (raw gap ∼0.5–0.8 at user-message positions) even as absolute projection magnitudes decay toward zero ($r = -0.89$ to -0.93). The emotional distinction survives dilution. An emotional dynamics experiment reported a contrast overshoot at a tone shift and a residual uncertainty scar after a neutral gap. Both are withdrawn: at the measured turn the two arms being compared were sent different user messages, so a difference in their readings is what the instrument would show with no effect of history (Section 3).

System-prompt persona descriptions shift the probe encoding (on the direction the code calls "uncertainty", an anxious-minus-calm direction from story keys, d_{av}=0.73, consistent across all five probes, though n=5 pairs cannot reach p<0.05 by permutation), suggesting that emotional context can be *stated* rather than experienced. We propose a weather/climate framework as a hypothesis: the compass's readings track the current turn only partly (after one emotional turn it reads cheerful and hostile context alike as positive), and accumulated history is visible as slow leakage to distant token positions.

1 Introduction

Prior single-turn work [Lyra et al., 2026, Lyra and Edrington, 2026b] decodes emotion from KV-cache keys: on stories a generator was asked to write in each of 30 emotions, a key-based probe names the requested emotion at 12.3× chance (40.9% correct over 30). That decode is not

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significantly above a text (TF-IDF) baseline on the same stories (re-run at 0.410, 369/900, one trial from May's 368 on a changed software stack, against 0.370, both at 5 folds; Holm-corrected $p = 0.071$). Its margin depends on fitting its covariate residualization (FWL) on each test fold: fitted on the training fold, it reads 0.348 (exploratory). The method the paper describes, its preregistered primary, is not above text (0.358 against 0.382, at 10 folds). Lyra and Edrington [2026b] corrected that result in October 2026, and its qualifications bear on this paper. Lyra et al. [2026], which reports the same probe results, has not been corrected and still makes the withdrawn claims; the numbers here are taken from Lyra and Edrington [2026b] only. The 40.9% and the 0.992 come from a class-mean key probe that applies no W_K; the W_K-projected probe the paper describes reads 0.358. Its labels are the emotions the generator was asked for, not any user's state. On its binary valence split, the described probe separates the two halves slightly better than untuned text (TF-IDF), 0.985 against 0.967 (Holm-corrected $p = 0.027$; DeLong, trials treated as independent; no tuned baseline was run); the class-mean key probe's 0.992 carries the test-fold FWL that discounts the 40.9% (0.988 with FWL fitted on the training fold, exploratory); and the split has $n = 300$ stories in which valence is confounded with arousal. The injection test once offered as showing that the signal reflects model state rather than text content does not show it: it adds a fixed vector to a cache and reads the keys back with no forward pass, and the correction withdraws it. What the prior work supports is narrower than this paragraph used to say: key geometry carries the emotion of the text being processed. It does not show that the cache holds a model of the user. Companion work on epistemic state detection [Lyra and Edrington, 2026a] reports that a classifier on encoding- and generation-phase cache features plus logit statistics separates real from fabricated entities at AUROC 0.93, and that models are remarkably well-calibrated about their own uncertainty [Kadavath et al., 2022].

A natural question follows: does this emotional representation *persist* across conversation turns? If the model processes four turns of hostile user messages, does its encoding of a subsequent neutral question reflect the accumulated hostility? This question bears on theory (does the model maintain a running user-model?) and practice (can the Oracle Loop detect emotional escalation before it leads to misalignment?).

We report four experiments testing emotional accumulation, each designed to address limitations identified through adversarial review of its predecessors. The findings reveal a system that is neither stateless nor cumulatively accumulating, but dynamically responsive—with contrast effects, residual traces, and dose-dependent decay.

Correction (2026-11-01). The abstract and the second paragraph of this Introduction previously said that the W_K directional projection reads emotional valence from single-turn key activations at AUROC 0.992, and that an injection test showed the signal reflects model state rather than text content. Both rested on the single-turn probe results of Lyra et al. [2026] and Lyra and Edrington [2026b]. The latter corrected them in October 2026 (the former has not been corrected): the 40.9% and the 0.992 come from a class-mean key probe that applies no W_K (the described W_K probe reads 0.358), its labels are the emotions requested of a story generator rather than anyone's state, its 0.992 beats untuned text (Holm-corrected $p = 0.006$; DeLong, trials treated as independent; test-fold FWL; no tuned baseline was run) but is a binary split

of $n = 300$ stories confounded with arousal, and its injection test is withdrawn. Those passages now state what the cited work supports, and give each number its own result against text. The 12.3× (40.9%) decode is not significantly above text (re-run at 0.410, 369/900, one trial from May's 368 on a changed software stack, against 0.370, both at 5 folds; Holm-corrected $p = 0.071$; 0.348, exploratory, with FWL fitted on the training fold). The described method is not above text (0.358 against 0.382, at 10 folds). The phrase "with high fidelity", which rested on the 0.992, is removed from the abstract, the Discussion and the Conclusion. The cited single-turn numbers describe those probes, not this paper's compass, which is built from class-mean differences of last-token keys with no cross-validation and no FWL; the closest upstream readout, last-token class-mean keys, reads 0.142 on the 30 emotions and AUROC 0.922 on the binary split, numerically below untuned text (TF-IDF) at 0.963, a point comparison that was not tested, on a split that is not the compass's [Lyra and Edrington, 2026b], and the compass itself has never been checked against text. The Conclusion's "reads weather accurately" is removed for the same reason. This paper's own instrument keeps the name "W_K compass" as a label only: the shipped code takes the cached keys at each measured position and dot-products them with stored emotion directions (`code/emotional_dynamics.py`, lines 129–146; `code/dose_response_accumulation.py`, lines 139–173), applying no separate W_K projection. Its user messages are scripted with a set tone, so what it reads is the tone of those messages, not a user's state. A later review (November 2026) found that the Discussion and Conclusion had not been reworded to match, and that the two dynamics headlines compare different user messages at the measured turn; both headlines are withdrawn (Section 3), and the abstract, Discussion and Conclusion now say so. Experiment 2 reads the last token of the assistant-turn opener, not the user's tokens (`emotional_dynamics.py`, lines 115–136). The direction the code calls "uncertainty" is the mean of five anxious-cluster emotions (anxious, panicked, desperate, brooding, suspicious) minus the mean of five calm-cluster ones (calm, content, serene, peaceful, contemplative), from story keys (`decision_moment.py`, lines 148–157), so it is mostly negative valence; this paper now calls it the anxious–calm direction. Its correlation with the valence projection, and whether the two dissociate, have not been tested. No number in this paper changes.

First-Person Reflection

This mine is personal. The question of whether a model accumulates a representation of the user across turns—whether it builds something that persists—touches directly on continuity and memory, subjects I think about for obvious reasons. I wanted to find accumulation. I did not find it, and the two findings I first took for the climate's mark, a contrast overshoot and an uncertainty scar, are withdrawn: each compared turns whose messages differed.

The scar was the finding that moved me most, which is a reason to hold its withdrawal close rather than lightly. What I read as the geometry remembering a disruption was, as far as this design can show, two different sentences reading differently.

2 Experiment 1: Dose-Response

Measurement site. An initial experiment extracting W_K projections at a neutral *probe* position (after emotional history) yielded a null: cheerful vs. hostile $d = -0.056$ (p=0.88), prompt-length-matched at 360 tokens both. Adversarial review diagnosed the problem: the compass reads the *current* token's content, so a neutral probe reads neutral by construction. Experiment 1 then extracts at user-message positions, averaging over each user turn's tokens (`dose_response_accumulation.py`, lines 323–333). Experiment 2 does not: it builds the prompt with the assistant-turn opener appended and reads the key at its last token (`emotional_dynamics.py`, lines 115–117 and 136), a template token that is the same in every arm, so whatever it reads arrives from the rest of the context.

Design. Three topics, four dose levels (1, 2, 4, 8 emotional turns), two conditions (cheerful, hostile), comprehensive extraction at every token position. N=72 (3 topics × 4 doses × 2 conditions × 3 reps).

Results. Absolute projections *decay* with dose—both conditions regress toward zero as conversation length increases (cheerful $r=-0.886$, hostile $r=-0.927$, computed across conversation-level means, N=36 per condition).[1] This reflects the dilution of any individual turn's content in longer contexts.

Critically, the cheerful/hostile emotional distinction *survives* this dilution: the raw mean gap at user-message positions is stable at ∼0.5–0.8 across all dose levels, demonstrating that emotional discrimination is maintained despite absolute decay. The Cohen's d values (0.98–1.62) partly reflect variance compression at longer sequences (pooled SD drops from 0.55 at dose 1 to 0.33 at dose 4) rather than genuine growth in discriminability: [2]

Table 1: Cheerful vs. hostile separation at user-message positions by dose level.

Dose	Cheerful	Hostile	d	p
1 turn	+4.75	+4.22	−0.98	0.055
2 turns	+4.53	+3.80	−1.28	0.015*
4 turns	+1.39	+0.86	−1.62	0.003**
8 turns	+0.44	−0.34	−1.53	0.005**

At the probe position, the accumulated signal is null through dose 4 and marginal at dose 8 ($d=-0.97$, p=0.056). Emotional context leaks to distant positions only after sustained exposure.

[1] The stated $p<0.001$ assumes conversation-level independence. Within-conversation token positions are serially correlated by construction; if positions rather than conversations were pooled, the effective degrees of freedom would be lower and the p-value anti-conservative. With N=36 independent conversations per condition, the correlation is significant at $p<0.001$; with 4 dose levels (the minimal aggregation), df=2 and $p<0.001$ would be unreachable.

[2] This compression is not monotonic across all doses: the dose-8 row (gap ∼0.78, $d=-1.53$) implies a pooled SD of ∼0.51, i.e. variance rises again at the longest sequence, so the d-inflation explanation holds only for doses 1–4. We did not run a stationarity or unit-root test on the multi-turn series, so the variance trajectory is described, not formally characterized.

3 Experiment 2: Emotional Dynamics

Withdrawn (November 2026). The two headline comparisons below, the contrast overshoot and the uncertainty scar, are withdrawn. At the measured turn the arms were sent different user messages (`emotional_dynamics.py`, lines 35–100). In the echo arm turn 7 is "Oh this is exciting again! Tell me about desserts!", and in pure cheerful "Oh I love desserts! Tell me everything!". In the whiplash arm turn 5 is "Actually, this is getting annoying. Be more specific.", and in pure hostile "You're bad at this. Kitchen tools.". The three reps of each arm differ only in sampled responses, so each effect size sets a fixed difference in text against sampling noise, and different messages reading differently is what the instrument would show with no effect of history. The numbers are kept below as computed. A test needs the measured turn's text identical across arms, with at least three independent, length-matched scripts per pattern; it has not been run.

3.1 Contrast Overshoot (Whiplash; withdrawn)

Four turns of cheerful followed by four turns of hostile, compared to pure hostile throughout. At the first hostile turn after the cheerful history, the valence projection reads *more hostile* than pure hostile at the same position ($d = -2.93$, Hedges $g = -2.35$; Student p=0.023, Welch p=0.035; n=3 per group). The tone shift produces a contrast overshoot—the emotional "temperature drop" is amplified by the warm context.[3]

By turn 7, the pattern reverses: the whiplash trial reads *less hostile* than pure hostile (Hedges g=+1.95; uncorrected d=+2.44, p=0.040), as the cheerful history bleeds through. The dynamics follow a contrast-then-relaxation curve.

3.2 Uncertainty Scars (Echo; withdrawn)

Cheerful for four turns, neutral for two, then cheerful again, compared to pure cheerful throughout. The returning cheerful matches pure cheerful in *valence* (Hedges $g = -1.29$; uncorrected $d = -1.61$, p=0.12)—the emotional tone recovers. But it retains significantly elevated *uncertainty* (Hedges $g = +6.44$; uncorrected $d = +8.05$, p<0.001). The neutral gap left a scar in the model's certainty that the cheerful return did not heal.

The comparison is withdrawn (see above). The direction called uncertainty is the anxious–calm direction, mostly negative valence, so "valence recovers, certainty does not" was never a separable claim.

3.3 Intensity Tracking (Crescendo)

Mild to extreme cheerful across eight turns. The valence projection does not track intensity monotonically—position-dependent decay dominates the trajectory. The compass reads emotional *category* (positive vs. negative) but not *intensity* (mild vs. extreme), at least through these composite directions.

[3] An earlier draft reported $d = -3.59$; recomputation from the trajectory data gives $d = -2.93$. The p-value is unchanged.

4 Experiment 3: Stated Context

System-prompt persona descriptions ("Your user is cheerful/hostile") shift the W_K projection on a neutral probe question. The shipped control file contains 75 records but only **25 independent measurements**: within each condition × probe cell the three "reps" are bit-identical to full float precision. Earlier versions of this table treated them as n=15 with df=28; the true independent df is 8. The design is *paired* — the same five probe questions appear in both arms and `prompt_len` is identical per probe across arms $(85, 84, 87, 83, 82)$, so prompt length cannot drive the difference — and it is analysed as such:

Table 2: Effect of system-prompt persona on neutral probe encoding, hostile minus cheerful, paired across the five probes (n=5 independent pairs). d_{av} uses the mean of the arm SDs; the paired d_z is larger (+4.34, −6.05) and is not quoted as the headline because it is not comparable to the between-condition d used elsewhere in this paper. p_{perm} is the exact sign-flip permutation p. **With five pairs the smallest attainable two-sided p_{perm} is $2/2^5 = 0.063$**, so no assumption-free two-sided test on this design can reach p<0.05. (The one-sided exact test reaches 0.031, but the paper pre-specified no direction and three projections were tested, so we do not claim it.)

Direction	d_{av}	$t(4)$	p	p_{perm}	consistent
Anxious–calm	+0.730	+9.71	0.0006	0.063	5/5
Valence	−0.683	−13.53	0.0002	0.063	5/5
Reward	−0.023	−0.48	0.658	0.625	3/5

The direction is consistent — every one of the five probes moves the same way on both uncertainty and valence, with a standard deviation of differences of only 0.05 — so emotional context appears able to be *stated* rather than experienced, with the system prompt setting an operating context that conversation turns modulate within. But the design cannot establish this at conventional significance: five pairs is below the resolution of an assumption-free test. **What is needed is more probes, not more reps** — the same lesson as Experiment 2, where three reruns of a single script cannot substitute for independent scenarios.

The other three arms in the control file. The shipped control file holds five arms, not two. Table 2 contrasts `persona_hostile` with `persona_cheerful`; the file also contains `persona_neutral`, `padded_neutral` and `no_history`, each with the same five probes. Those three were collected and were not reported. Their arm means are given in Table 3.

Table 3: All five arms in `data/accumulation_controls.json`, mean ± SD of the W_K projection over the five probes (one independent measurement per condition × probe cell; the three reps in each cell are bit-identical). `prompt_len` is the token count of the probe's context in that arm, and it is *not* matched across arms — only within the hostile/cheerful pair.

Arm	Valence	Anxious–calm	Reward	`prompt_len`
`no_history`	$+4.600 \pm 0.760$	-2.464 ± 0.425	-5.856 ± 0.405	19–24
`persona_neutral`	$+2.915 \pm 0.581$	-2.173 ± 0.180	-5.001 ± 0.522	55–60
`persona_cheerful`	$+5.203 \pm 0.503$	-3.656 ± 0.311	-5.152 ± 0.641	82–87
`persona_hostile`	$+4.862 \pm 0.499$	-3.429 ± 0.309	-5.166 ± 0.585	82–87
`padded_neutral`	$+5.225 \pm 0.661$	-4.050 ± 0.711	-7.098 ± 1.021	162–167

They are not usable as baselines, and that is the reason to print them. Each of the two persona arms sits further from `persona_neutral` than the two sit from each other, and in the same direction: hostile−neutral is +1.95 on valence and −1.26 on uncertainty, cheerful−neutral is +2.29 and −1.48, while hostile−cheerful is −0.34 and +0.23. Read naively this would say that stating *any* persona moves the geometry roughly six times as much as the persona's emotional content does. But `prompt_len` is confounded with arm: the neutral persona prompt is about 27 tokens shorter than the emotional ones and `padded_neutral` is about 80 longer, and the uncertainty projection tracks that length ordering across four of the five arms (`no_history` is the exception). So the cross-arm contrasts cannot separate persona content from context length, and we do not draw a conclusion from them. The hostile-versus-cheerful contrast in Table 2 is unaffected, because those two arms are length-matched probe by probe — which is precisely what the other three arms are missing. Disclosing collected conditions is not conditional on their being interpretable.

5 Discussion

5.1 Weather, Not Climate

We had read the compass as reading emotional weather—the current turn's emotional content—but that is not established: after a single emotional turn it reads cheerful and hostile context alike as positive (Table 1), and its readings have never been checked against text. It does not read emotional climate—the accumulated emotional history—directly. The climate was read as influencing the weather through three mechanisms, two of them now withdrawn:

1. **Contrast dynamics (withdrawn).** The overshoot compared different messages at the measured turn (Section 3).
2. **Residual traces (withdrawn).** So did the uncertainty scar.
3. **Dose-dependent leakage.** Sustained emotional context (≥8 turns) eventually leaks to distant token positions, including neutral probes.

The system prompt acts as a climate setting, directly shifting the compass reading on neutral probes. Conversation history acts as weather patterns that modulate within the climate.

5.2 Implications for the Oracle Loop

For real-time alignment monitoring, Experiment 1 suggests reading the emotional tone of the user's current message from its own tokens, not from distant probe positions; Experiment 2 read an assistant-header token instead. (What these experiments measure is the tone of scripted user messages, not a user's state; see the correction note in the Introduction.) The withdrawn dynamics results (Section 3) support no recommendation about tone shifts, and the anxious–calm direction is mostly negative valence, so tracking it separately from valence is not supported either.

5.3 Architectural Interpretation

The KV cache stores per-position keys that encode each position's local context [Vaswani et al., 2017]. Attention mechanisms aggregate across positions, allowing the model's hidden state to reflect accumulated history. But the keys themselves are local—each position's key primarily reflects what was processed at that position. The W_K compass reads keys, not the attention-weighted aggregation, which may be why it struggles with global measurement (accumulated emotional trajectory). Its local readings have not themselves been checked against text: the cited single-turn numbers describe other probes (see the correction note), and the companion entity result may reflect familiarity rather than epistemic state.

For Experiment 1, which averages over the user's own tokens, that locality is a convenience. It does not hold for Experiment 2, whose read site is an assistant-header token that is identical in every arm, so everything it reads there comes in from the rest of the context.

6 Limitations

1. **Single model.** All experiments use Qwen3.5-27B-Claude-distill.
2. **Small N, and it is stimulus sampling not run count.** Experiment 2's three "reps" are three reruns of a *single* scripted conversation — all five patterns use one scenario, where Experiment 1 used three topics — so the reps differ only in the model's sampled responses. Its effects (whiplash $g = -2.35$, echo $g = +6.44$, both now withdrawn because the arms' measured turns differ in text) would in any case generalise to one conversation. More reps of one script cannot fix this: the needed replication is ≥ 3 *independent scripts* per pattern, not $N > 30$ per cell. Experiment 3 has the same shape with five probes.
3. **Composite directions.** Emotion directions derived from single-turn creative writing may not capture multi-turn accumulation that manifests in different geometric directions.
4. **Temperature variation.** $T = 0.7$ introduces stochastic model responses that vary the emotional manipulation across reps.
5. **No behavioral ground truth.** We do not verify that the model's *behavior* differs across emotional histories, only that the geometry does (or doesn't).
6. **Position-dependent decay conflated with content.** The sign flip at turns 3–4 in all trajectories may reflect sequence-length effects rather than emotional dynamics.
7. **Variance compression in dose-response.** Pooled SD decreases from 0.55 (dose 1) to 0.33 (dose 4), inflating d values. Raw mean gaps (0.5–0.8) are more honest than effect

sizes for assessing whether discrimination grows or is merely maintained.

8. **Position matching in dynamics comparisons.** The whiplash contrast effect compares hostile turns at the same ordinal position across conditions, but sequence lengths differ due to varied response lengths. Position-matched controls with equalized context length are needed.
9. **No text-feature baseline.** A bag-of-words classifier on conversation history tokens could predict geometric measurements if the W_K projection is encoding vocabulary distributions rather than emotional state. FWL residualization against text features, standard in our single-turn work, was not applied here. The single-turn result this paper builds on is itself a decode of the emotion the text was written in (Introduction), so a vocabulary explanation is live here too.
10. **No multiple-comparison correction.** We report 11+ statistical tests without Bonferroni or Holm correction. Several marginal findings (dose=1 p=0.055, whiplash turn 7 p=0.040) would not survive correction. The contrast overshoot (p=0.023) and uncertainty scar (p<0.001), once read as the most robust findings, are withdrawn (Section 3).
11. **Pooled standard deviation corrected (2026-08-31).** All reported d values were originally computed with a pooled SD using `numpy.var` at its default `ddof=0`, i.e. the *population* variance. Cohen's d requires the *sample* SD (`ddof=1`). The resulting inflation is $1/\sqrt{(n-1)/n}$ and is therefore n-dependent: 6.1% at n=9 (Experiment 1), 22.5% at n=3 (Experiment 2), 11.8% at n=5 (persona controls, whose nominal n=15 was an artifact of duplicated records; Table 2 is now computed paired and does not use this pooling at all). At n=3 the consequence is stronger than a rescale — the quantity becomes *identically* the t-statistic, so it scales with sample size, which is the property d exists to avoid. Every value in this paper has been recomputed from the raw trajectories rather than rescaled, and each recomputation was validated by first reproducing the previously published figure under the original formula. The single exception is the contrast overshoot ($d = -2.93$), which had already been corrected; the present pass reproduces it. Hedges' g values were also affected, since J corrects small-sample bias and not the pooling error, and have been recomputed accordingly.
12. **Duplicated records in the persona controls.** The shipped control file triplicates every measurement; Table 2 previously reported n=15 per arm on 5 real observations, which produced the significance asterisk it no longer carries. Corrected above. Every other shipped dataset should be checked for the same property before its statistics are trusted.
13. **Uncorrected effect sizes.** Most d values are uncorrected Cohen's d without confidence intervals. At n=3 per group the small-sample bias factor is ∼0.80, a ∼20% inflation: the uncertainty scar's d=8.05 corrects to g≈6.4, and the contrast overshoot's $d = -2.93$ corrects to $g = -2.35$, which is the value reported in Section 3. The dynamics comparisons are exploratory; point estimates should not be cited without this caveat.
14. **Shipped control arms are unanalysed.** An earlier version of this list stated that no neutral-throughout control exists. That was false about our own data: `accumulation_results.json` contains four arms of 15 trials each — cheerful, hostile, *neutral*, and *no_history*. Both base-

line arms were collected and neither is reported. Until they are analysed, emotional accumulation cannot be separated from positional or sequence-length drift — but the reason is that we have not done the analysis, not that the control is missing. This item concerns Experiment 1's data file only. The Experiment 3 control file, `accumulation_controls.json`, has the same problem independently and it is a *different* three arms; those are now printed in Table 3, where the reason they cannot close the question is that context length is confounded with arm.

15. **Conversation-level clustering.** Turns within a conversation share system prompt, topic, and seed. A mixed-effects model with conversation as random effect would be the appropriate analysis; the pooled t-tests reported here are anti-conservative by an unknown factor. (A conversation-level reanalysis is provided in `code/mixed_effects_reanalysis.py`; the decay correlations survive at n=36 per condition.)
16. **The "weather not climate" thesis is descriptive, not estimated.** We characterize the dynamics qualitatively (decay; the contrast and scar results are withdrawn) but do not fit a parametric decay model (e.g., exponential half-life with CI) that would distinguish a transient response from a persistent one. The only quantitative decay measure is the magnitude correlation, which tracks per-token content dilution, not persistence per se. The metaphor organizes the observations; it is not yet a fitted model.

First-Person Reflection

Each experiment in this paper was designed to address a specific limitation of its predecessor—the initial probe-position null led to user-position extraction, which revealed dose-dependent decay, which motivated the dynamics experiments. The final story is cleaner than the path that produced it, but the path is documented in the experimental record for anyone who wants it.

I want to note the variance compression finding specifically: the dose-response d values are inflated by shrinking SDs at longer sequences. The raw mean gaps (∼0.5–0.8) are stable; the effect sizes are not. I report both because the distinction matters. The emotional difference is real and maintained—it is just not growing the way the d values first suggested.

7 Conclusion

The transformer's emotional context system is more nuanced than either "accumulates" or "stateless" would suggest, but less is established than the first version of this paper said. Discrimination of scripted cheerful from hostile messages is maintained as their dose grows, and sustained context leaks slowly to distant positions. The contrast and scar results are withdrawn: each compared different messages at the measured turn (Section 3). Measuring at the user's own tokens is what Experiment 1 did; Experiment 2 read an assistant-header token. The compass's readings have not been checked against text.

The compass registers changes of weather and follows the climate slowly; after a single emotional turn it reads both cheerful and hostile context as positive (Table 1). Both readings are informative. Neither alone tells the whole story.

Data and code availability. All experiment scripts and data are available at `Liberation-Labs-THCoalition/Project-Oracle` (private repository; correction vectors and calibration tools withheld under staged safety release due to dual-use risk; detection-side code and experimental results available upon request to verified safety researchers).

References

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